

Covered: Continuous functions are Riemann Integrable.

1 Riemann Integration (contd.)

Let $Q \subseteq \mathbb{R}^n$ be a rectangle. For a partition P of Q , and for each subrectangle $R \in P$, let

$$m_R := \inf_{x \in R} f(x), \quad M_R := \sup_{x \in R} f(x).$$

Example 1.1 (constant function). Let $f : Q \rightarrow \mathbb{R}$ defined $f(x) = c$ be constant. We can partition Q into subrectangles R such that $m_R = M_R = c$. Hence

$$\underline{\int}_Q f = \sup_P \sum_{R \in P} m_R \operatorname{vol}(R) = c \sum_{R \in P} \operatorname{vol}(R) = c \operatorname{vol}(Q),$$

while

$$\overline{\int}_Q f = \inf_P \sum_{R \in P} M_R \operatorname{vol}(R) = c \sum_{R \in P} \operatorname{vol}(R) = c \operatorname{vol}(Q).$$

Therefore f is Riemann integrable and

$$\int_Q f = c \operatorname{vol}(Q).$$

Example 1.2 (Dirichlet function). Let

$$f : Q \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} 1, & \pi_i(x) \in \mathbb{Q} \quad \forall i = 1, \dots, n, \\ 0, & \text{otherwise,} \end{cases}$$

where $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ denotes the i -th coordinate projection.

Then every subrectangle $R \subseteq Q$ contains points of $\mathbb{Q}^n$ as well as points outside $\mathbb{Q}^n$. Hence

$$m_R = 0, \quad M_R = 1$$

for every subrectangle R .

Thus

$$\underline{\int}_Q f = \sup_P \sum_{R \in P} m_R \operatorname{vol}(R) = 0,$$

whereas

$$\overline{\int}_Q f = \inf_P \sum_{R \in P} M_R \operatorname{vol}(R) = \inf_P \sum_{R \in P} \operatorname{vol}(R) = \operatorname{vol}(Q).$$

Therefore, if $\operatorname{vol}(Q) > 0$,

$$\underline{\int}_Q f \neq \overline{\int}_Q f,$$

so f is not Riemann integrable.

Theorem 1.3. Let C be a compact metric space and let $f : C \rightarrow \mathbb{R}$ be continuous. Then f is uniformly continuous.

Proof. Suppose, to the contrary, that f is not uniformly continuous. Then there exists $\epsilon_0 > 0$ such that for every $n \in \mathbb{N}$, there exist $x_n, y_n \in C$ satisfying

$$d(x_n, y_n) < \frac{1}{n}$$

but

$$|f(x_n) - f(y_n)| \geq \epsilon_0.$$

Since C is compact, the sequence (x_n) has a convergent subsequence $x_{n_k} \rightarrow x \in C$. Moreover,

$$d(x_{n_k}, y_{n_k}) < \frac{1}{n_k} \rightarrow 0,$$

hence $y_{n_k} \rightarrow x$. Since f is continuous,

$$f(x_{n_k}) \rightarrow f(x), \quad f(y_{n_k}) \rightarrow f(x).$$

Therefore

$$|f(x_{n_k}) - f(y_{n_k})| \rightarrow 0,$$

contradicting

$$|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon_0.$$

Hence f is uniformly continuous. $\square$

Lemma 1.4. Let $f : Q \rightarrow \mathbb{R}$, where Q is a rectangle. Then f is Riemann integrable if and only if, given $\epsilon > 0$, there exists a partition P of Q such that

$$U(P, f) - L(P, f) < \epsilon.$$

Proof. ($\Rightarrow$) We have f Riemann integrable, so

$$\int_Q f = \sup_P L(P, f) = \inf_P U(P, f) = \overline{\int_Q f}$$

Let $\epsilon > 0$. Then there exists a partition P_1 such that

$$\int_Q f - \frac{\epsilon}{2} < L(P_1, f)$$

and there exists a partition P_2 such that

$$\overline{\int_Q f} + \frac{\epsilon}{2} > U(P_2, f)$$

Let $P = P_1 \cup P_2$ be a common refinement of P_1 and P_2 . Since lower sums increase under refinement and upper sums decrease under refinement,

$$L(P, f) \geq L(P_1, f), \quad U(P, f) \leq U(P_2, f).$$

Hence

$$\int_Q f - \frac{\epsilon}{2} < L(P, f) \leq U(P, f) < \overline{\int_Q f} + \frac{\epsilon}{2},$$

and therefore

$$U(P, f) - L(P, f) < \epsilon.$$

($\Leftarrow$) Suppose that for every $\epsilon > 0$ there exists a partition P such that

$$U(P, f) - L(P, f) < \epsilon.$$

Since

$$L(P, f) \leq \int_Q f \leq \overline{\int_Q f} \leq U(P, f),$$

we obtain

$$0 \leq \overline{\int_Q f} - \int_Q f \leq U(P, f) - L(P, f) < \epsilon.$$

Since this holds for every $\epsilon > 0$,

$$\overline{\int_Q f} - \int_Q f = 0 \implies \boxed{\overline{\int_Q f} = \int_Q f}$$

Hence f is Riemann integrable. $\square$

Theorem 1.5. If $f : Q \rightarrow \mathbb{R}$ is continuous, where $Q \subseteq \mathbb{R}^n$ is a rectangle, then f is Riemann integrable.

Proof. Given $\epsilon > 0$, since f is uniformly continuous on the compact rectangle Q , there exists $\delta > 0$ such that

$$\|x - y\| < \delta \implies |f(x) - f(y)| < \frac{\epsilon}{\text{vol}(Q)}.$$

Choose a partition P of Q whose mesh satisfies $\|P\| < \delta$. For every rectangle $R \in P$,

$$M_R - m_R < \frac{\epsilon}{\text{vol}(Q)},$$

where

$$M_R = \sup_{x \in R} f(x), \quad m_R = \inf_{x \in R} f(x).$$

Therefore

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{R \in P} (M_R - m_R) \text{vol}(R) \\ &< \frac{\epsilon}{\text{vol}(Q)} \sum_{R \in P} \text{vol}(R) \\ &= \frac{\epsilon}{\text{vol}(Q)} \text{vol}(Q) = \epsilon. \end{aligned}$$

By the lemma, f is Riemann integrable. Choose the partition P so that, writing the coordinate endpoints as $t_i^{(j)}$,

$$\left| t_i^{(j)} - t_{i-1}^{(j)} \right| < \frac{\delta}{\sqrt{n}}, \quad \forall i, j.$$

Then, given any $x, y \in R$, where R is a subrectangle of P ,

$$|x_j - y_j| < \frac{\delta}{\sqrt{n}}, \quad j = 1, \dots, n,$$

and hence

$$\|x - y\|_2 = \left(\sum_{j=1}^n |x_j - y_j|^2 \right)^{1/2} < \left(n \frac{\delta^2}{n} \right)^{1/2} = \delta.$$

Therefore, for any $x, y \in R$,

$$|f(x) - f(y)| < \frac{\epsilon}{\text{vol}(Q)} \implies M_R - m_R < \frac{\epsilon}{\text{vol}(Q)}$$

and consequently

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{R \in P} (M_R - m_R) \text{vol}(R) \\ &< \frac{\epsilon}{\text{vol}(Q)} \sum_{R \in P} \text{vol}(R) \\ &= \frac{\epsilon}{\text{vol}(Q)} \text{vol}(Q) = \epsilon. \end{aligned}$$

Hence, by the preceding lemma, f is Riemann integrable. $\square$

Exercise 1.6. 1. Prove that if f is not continuous at finitely many points, then f is Riemann integrable.

2. Show by an example that if f is discontinuous at infinitely many points, it can still be Riemann integrable.